

Cover Letter

Dear Editors,

Please consider our manuscript, "External validation audit of peripheral blood transcriptomic signatures in Alzheimer's disease reveals cohort-dependent reproducibility and limited transferability", for publication as a Research Article in BMC Medical Genomics.

This study presents a staged validation audit of public peripheral-blood transcriptomic datasets in Alzheimer's disease. GSE63060 was used for discovery, GSE63061 for related-cohort validation, GSE85426 for independent cross-platform stress testing, and GSE140829 for large independent external validation. The central finding is that a discovery-derived AD blood transcriptional signal reproduces strongly in a related cohort but shows limited transferability in independent validation, with model AUC values remaining below the prespecified study gate of 0.70.

The manuscript is intentionally framed as a cautious external-validation study rather than as a diagnostic classifier success report. We believe this negative-leaning transportability result is relevant to BMC Medical Genomics because it addresses reproducibility, public genomic data reuse, and the evidence standards needed before blood transcriptomic signatures are advanced as biomarker claims.

All datasets analyzed in the study are publicly available from GEO. No new human participants were recruited, no new specimens were collected, and the study used de-identified public data only. The author declares no competing interests and no specific funding was received for this work.

This manuscript is not under consideration elsewhere. The author has approved the submitted version.

Sincerely,

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